# Kenneth Blake Vernon

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## Professional Appointments

**One-U Responsible AI Postdoctoral Fellow** | 2025-Present
Scientific Computing and Imaging Institute
School of Environment, Society, and Sustainability
University of Utah

**Post-Doctoral Associate** | 2022-2024
Center for Collaborative Synthesis in Archaeology, CU Boulder

**Assistant Director** | 2018-2022
University of Utah Archaeological Center

## Education

**University of Utah** | Salt Lake City, UT | PhD in Anthropology, 2022

**Northern Illinois University** | DeKalb, IL | MA in Philosophy, 2009

**University of Central Arkansas** | Conway, AR | BA in History and Philosophy, 2006

## Publications

### *Peer-reviewed journal articles*

**2026**

**A reconstruction of the Indigenous maize farming niche of Utah**

*Journal of Archaeological Science: Reports* 69: 105493 (2026)

Ishmael D. Medina, Kenneth B. Vernon, Weston C. McCool, Jerry D. Spangler, Brian F. Codding

doi 10.1016/j.jasrep.2025.105493

## 2025

### Targeted explorations of Pleistocene-Holocene transition archaeology on the Colorado Plateau in southern Utah

*Quaternary International* 746: 109992 (2025)

Abby S. Baka, Kenneth B. Vernon, Madeline E. Mackie, Simon Brewer, Jerry Spangler, Brian F. Codding, Lisbeth A. Louderback, Thomas H. Flanigan, Alexandra M. Greenwald

10.1016/j.quaint.2025.109992

### Drought, Population Pressure, and Inequality Drive Intergroup Conflict in the Precontact North American Southwest

*American Antiquity*: 1-19 (2025)

Weston C. McCool, Kenneth B. Vernon, Ishmael D. Medina, Joan Brenner Coltrain, Kurt M. Wilson, Brian F. Codding

10.1017/aaq.2025.10107

### Modeling the rise and demise of Classic Maya cities: Climate, conflict, and economies of scale

*Proceedings of the National Academy of Sciences* 122 (42): e2512325122 (2025)

Weston C. McCool, Brian F. Codding, Bridgette Degnan, Claire E. Ebert, Emily S. Johnson, Kenneth Blake Vernon, Kurt M. Wilson, Timothy Beach, Keith M. Prufer, Douglas J. Kennett

10.1073/pnas.2512325122

### Environmental drivers of wealth inequality among Ancestral Puebloan farmers in Bears Ears National Monument

*American Antiquity*: 1-7 (2025)

Roxanne Lois F. Lamson, Kurt M. Wilson, Hayley T. Kievman, Kenneth Blake Vernon, Peter M. Yaworsky, Brian F. Codding

10.1017/aaq.2025.8

## 2024

### Trade-off between market and ecosystem services drives settlement decisions among smallholder ranchers in Baja California Sur, Mexico

*Sustainability Science* 19 (4): 1377-1390 (2024)

Kenneth B. Vernon, Brian F. Codding, Simon C. Brewer, Shane J. Macfarlan

10.1007/s11625-024-01491-9

### The Fremont frontier: Living at the margins of maize farming

*American Antiquity* 89 (3): 440-458 (2024)

Kenneth B. Vernon, Peter M. Yaworsky, Weston McCool, Jerry D. Spangler, Simon Brewer, Brian F. Codding

10.1017/aaq.2024.22

### A method for defining dispersed community territories

*Journal of Archaeological Science* 170: 106048 (2024)

Kenneth B. Vernon, Scott G. Ortman

10.1016/j.jas.2024.106048

### The Goldilocks Zone for maize agriculture and the settlement and abandonment of the West Tavaputs Plateau

*Quaternary International* 689-690: 30-42 (2024)

Peter M. Yaworsky, Kenneth B. Vernon, Weston C. McCool, Isaac A. Hart, Jerry Spangler, Brian F. Codding

10.1016/j.quaint.2023.12.003

### Can we reliably detect adaptive responses of hunter-gatherers to past climate change? Examining the impact of Mid-Holocene drought on Archaic settlement in the Basin-Plateau Region of North America

*Quaternary International* 689-690: 5-15 (2024)

Brian F. Codding, Heidi Roberts, William Eckerle, Simon C. Brewer, Ishmael D. Medina, Kenneth B. Vernon, Jerry S. Spangler

10.1016/j.quaint.2023.06.014

## 2022

### Ecological constraints on violence avoidance tactics in the Prehispanic Central Andes

*Environmental Archaeology*: 1-14 (2022)

Weston C. McCool, Kurt M. Wilson, Kenneth B. Vernon

10.1080/14614103.2022.2137652

### Socioecological dynamics structuring the spread of farming in the North American basin-plateau region

*Environmental Archaeology* 27 (4): 434-446 (2022)

Brian F. Codding, Joan Brenner Coltrain, Lisbeth Louderback, Kenneth Blake Vernon, Kate E. Magargal, Peter M. Yaworsky, Erick Robinson, Simon C. Brewer, Jerry D. Spangler

10.1080/14614103.2021.1927480

### Decomposing habitat suitability across the forager to farmer transition

*Environmental Archaeology* 27 (4): 420-433 (2022)

Kenneth B. Vernon, Peter M. Yaworsky, Jerry Spangler, Simon Brewer, Brian F. Codding

10.1080/14614103.2020.1746880

### Subsistence strategy mediates ecological drivers of human violence

*PLOS ONE* 17 (5): e0268257 (2022)

Weston C. McCool, Kenneth B. Vernon, Peter M. Yaworsky, Brian F. Codding

10.1371/journal.pone.0268257

### Climate change–induced population pressure drives high rates of lethal violence in the Prehispanic central Andes

*Proceedings of the National Academy of Sciences* 119 (17): e2117556119 (2022)

Weston C. McCool, Brian F. Codding, Kenneth B. Vernon, Kurt M. Wilson, Peter M. Yaworsky, Norbert Marwan, Douglas J. Kennett

10.1073/pnas.2117556119

## 2021

### Plant species richness at archaeological sites suggests ecological legacy of Indigenous subsistence on the Colorado Plateau

*Proceedings of the National Academy of Sciences* 118 (21): e2025047118 (2021)

Bruce M. Pavlik, Lisbeth A. Louderback, Kenneth B. Vernon, Peter M. Yaworsky, Cynthia Wilson, Arnold Clifford, Brian F. Codding

10.1073/pnas.2025047118

## 2020

### Marriage dynamics in old Lower California: Ecological constraints and reproductive value in an arid peninsular frontier

*Biodemography and Social Biology* 65 (2): 156-171 (2020)

Shane J. Macfarlan, Ryan Schacht, Caroline Foley, Sydney Cahoon, Grace Osusky, Kenneth B. Vernon, Evan Tayler, Celeste Henrickson, Eric Schniter

10.1080/19485565.2020.1728685

### Advancing predictive modeling in archaeology: An evaluation of regression and machine learning methods on the Grand Staircase-Escalante National Monument

*PLOS ONE* 15 (10): e0239424 (2020)

Peter M. Yaworsky, Kenneth B. Vernon, Jerry D. Spangler, Simon C. Brewer, Brian F. Codding

10.1371/journal.pone.0239424

**2017**

### The ecology of population dispersal: Modeling alternative Basin-Plateau foraging strategies to explain the Numic expansion

*American Journal of Human Biology* 29 (4): e23000 (2017)

Kate E. Magargal, Ashley K. Parker, Kenneth Blake Vernon, Will Rath, Brian F. Codding

10.1002/ajhb.23000

## *Manuscripts*

### Migration and environmental learning in interstitial areas: An example from the Northern Rio Puerco Valley, New Mexico

*Open Archaeology* (accepted)

Kenneth B. Vernon, Scott G. Ortman, Donald A. Shiffler

### Responsible artificial intelligence and the science of human-environment interactions

*Environmental Data Science* (in prep)

Kenneth B. Vernon, Brian F. Codding, Simon C. Brewer

### The evolution of a labor market

(in prep)

Kenneth B. Vernon, Kelsey Carlston, Weston C. McCool, Kurt M. Wilson, Brian F. Codding

### One theory, many models of habitat suitability

(in prep)

Kenneth B. Vernon

### The impact of subsistence change on demography: Theory-informed Approximate Bayesian Computation reveals 4-fold increase in carrying capacity following economic intensification

*Journal of Archaeological Science* (under review)

Kurt M. Wilson, Brian F. Codding, Weston C. McCool, Ishmael Medina, Kenneth B. Vernon, Simon C. Brewer

### Explaining PaleoIndian settlement in the Intermountain West

*American Antiquity* (in prep)

Kenneth B. Vernon, Kate E. Magargal, David Zeanah, D. Craig Young, Robert G. Elston, Simon C. Brewer, Brian F. Codding

### NetC14: A Gaussian mixture neural network for calibrating and summarizing radiocarbon dates

*Journal of Archaeological Science* (in prep)

Kenneth B. Vernon, Brian F. Codding, Simon C. Brewer

### Socio-economic status before we measured it: A machine learning approach to imputing income and education in the US Census before 1950

(in prep)

Kelsey Carlston, Kenneth B. Vernon

### Predicting PHT archaeology on the northwest Colorado Plateau: A predictive model for determining targeted survey areas

*Journal of Archaeological Science: Reports* (in prep)

Abby S. Baka, Kenneth B. Vernon, Simon C. Brewer, Brian F. Codding, Jerry D. Spangler, Alexandra M. Greenwald

## *Book reviews*

#### 2020

### Review of Oshara Revisited: The Archaic Period in Northern New Mexico by Nicholas Chapin

*Journal of Anthropological Research* 76 (1): 138–140 (2020)

Kenneth B. Vernon, Brian F. Codding

10.1086/706946

## *Technical reports*

#### 2022

### Archaeological Survey of Cottonwood Wash, San Rafael Desert, Emery County, Utah

Report submitted to the Bureau of Land Management, Price Field Office, Price, Utah; and State of Utah School and Institutional Trust Lands Administration, Salt Lake City, Utah. Prepared by the University of Utah Archaeological Center and the Colorado Plateau Archaeological Alliance. BLM Cooperative Agreement L20AC00267. United States Department of Interior Permit No. 21UT85115. State of Utah Antiquities Project No. U22UA031. University of Utah Archaeological Center Project No. UUAC22052.

Ishmael Medina, Kenneth B. Vernon, Kate E. Magargal, Weston C. McCool, Jerry D. Spangler, Brian F. Codding

#### 2021

### Limited Test Excavations at Red Reef Shelter (42WS519), Red Cliffs National Conservation Area, Washington County, Utah

Report submitted to the Bureau of Land Management, St. George, Utah.

Jerry D. Spangler, Matthew Zweifel, Heidi Roberts, Kenneth B. Vernon, Isaac Hart, Brian F. Codding, Anne Lawlor

### 2019

**Hisatsinom of the high plateaus: a Class I overview of prehistoric cultural resources in Grand Staircase-Escalante National Monument**

Report submitted to the Bureau of Land Management, Salt Lake City, Utah.

Jerry D. Spangler, Peter M. Yaworsky, Kenneth B. Vernon, Brian F. Codding

### 2018

**Preliminary Archaeological Investigations at the Jack Kirk Site, Mariposa Creek, California**

Report submitted to the North Fork Mono Tribe, Mariposa Creek, California.

Brian F. Codding, Kate E. Magargal, Kenneth B. Vernon, Paul E. Allgaier, Jess Valdez, Ron Goode

**The Archaeology of Molen Reef, Emery County, Utah**

Report submitted to the Southern Utah Wilderness Alliance, Salt Lake City, Utah.

Kenneth B. Vernon, Peter M. Yaworsky, Jerry D. Spangler

## Grants, Fellowships, and Awards

### 2025

**extendr: Community-focused development for scientific computing with Rust and R**

ISC Grant, R Consortium.

Kenneth B. Vernon, Josiah Parry, Mossa M. Reimert

### 2022

**Archaeological Survey of Cottonwood Wash, San Rafael Desert, Emery County, Utah**

Cooperative Agreement, Bureau of Land Management.

Brian F. Codding, Jerry D. Spangler

### 2020

**Advancing Stewardship with Theory and Models**

David C. Williams Memorial Graduate Fellowship, University of Utah College of Social and Behavioral Sciences.

Kenneth B. Vernon

**Developing a Modular Online Introduction to the R Environment**

Cross-College Training Grant, University of Utah College of Social and Behavioral Sciences.

Simon Brewer, Brian F. Codding, Kenneth B. Vernon

## 2019

### Advancing Stewardship with Theory and Models

David C. Williams Memorial Graduate Fellowship, University of Utah College of Social and Behavioral Sciences.

Kenneth B. Vernon

### Graduate Research

Earl and Elies Skidmore Endowed Graduate Fellowship, University of Utah College of Social and Behavioral Sciences.

Kenneth B. Vernon

## 2018

### Advancing Stewardship with Theory and Models

David C. Williams Memorial Graduate Fellowship, University of Utah College of Social and Behavioral Sciences.

Kenneth B. Vernon

### Why Were Prehistoric Agricultural Societies in Utah Unable to Adapt to the Medieval Megadroughts?

Interdisciplinary Faculty Research Seed Grant, University of Utah Global Change and Sustainability Center.

Brian F. Codding, Simon Brewer

## 2017

### Archaeological Sites as Endangered Species: Using Next Generation Models to Predict and Protect Cultural Properties on the Grand Staircase-Escalante National Monument

Grand Staircase-Escalante National Monument Cooperative Agreement, Bureau of Land Management.

Brian F. Codding

### Human Plant Use in Red Butte Canyon

Research Grant, University of Utah Global Change and Sustainability Center.

Kenneth B. Vernon

### Human Plant Use in Red Butte Canyon

Research Grant, Friends of Red Butte.

Kenneth B. Vernon

# Presentations

## 2025

### Working with REST APIs to access spatial data

Spatial Utah Data Science (SUDS) Meeting, Salt Lake City, UT.

Kenneth B. Vernon

### Indirect dating with Mixture Density Networks

Utah Center for Data Science, Data Science & AI Lecture Series, Salt Lake City, UT.

Kenneth B. Vernon, Brian F Codding, Scott G. Ortman, Simon C. Brewer

### Promoting water-use efficiency with responsible AI

One-U Responsible AI Annual Symposium, Salt Lake City, UT.

Kenneth B. Vernon

### Indirect dating with Mixture Density Networks

90th Annual Meeting of the Society for American Archaeology, Denver, CO.

Kenneth B. Vernon, Brian F Codding, Scott G. Ortman, Simon C. Brewer

## 2024

### The network ecology of settlement persistence

30th Annual Meeting of the European Archaeological Association, Rome, Italy.

Kenneth B. Vernon, Simon C. Brewer, Brian F. Codding, Scott G. Ortman

### The network ecology of settlement persistence

workshop 'Advances in modelling past human ecosystems', Cologne, Germany.

Kenneth B. Vernon, Simon C. Brewer, Brian F. Codding, Scott G. Ortman

### The socio-ecology of community centers

89th Annual Meeting of the Society for American Archaeology, New Orleans, LA.

Kenneth B. Vernon, Scott G. Ortman

## 2023

### The ecology of agglomeration and the rise of Chaco Great Houses

88th Annual Meeting of the Society for American Archaeology, Portland, OR.

Kenneth B. Vernon, Weston C. McCool, Simon C. Brewer, Brian F. Codding, Scott G. Ortman

### Exploring the dynamics of late precontact obsidian procurement and exchange networks across the US Southwest and Northwest Mexico using cyberSW

The Southwest Symposium Archaeological Conference, Santa Fe, NM.

Sarah Oas, Kenneth B. Vernon, Matthew A. Peeples, Barbara J. Mills, Joshua Watts, Jeffery J. Clark, Steven Shackley, Scott G. Ortman

### How do I work this? this R project

Spatial Utah Data Science (SUDS) Meeting, Salt Lake City, UT.

Kenneth B. Vernon

### A core-periphery approach to defining community boundaries

29th Annual Meeting of the European Archaeological Association, Belfast, Northern Ireland.

Kenneth B. Vernon, Scott G. Ortman

## 2022

### The Fremont frontier: Living at the margins of maize farming

87th Annual Meeting of the Society for American Archaeology, Chicago, IL.

Kenneth B. Vernon, Jerry D. Spangler, Weston C. McCool, Peter M. Yaworsky, Simon C. Brewer, Brian F. Codding

## 2021

### Explaining PaleoIndian settlement in the Intermountain West: A regression adjustment approach

86th Annual Meeting of the Society for American Archaeology, San Francisco, CA.

Kenneth B. Vernon, David Zeanah, D. Craig Young, Robert G. Elston, Brian F. Codding

## 2019

### Decomposing habitat suitability with theory-driven machine-learning

84th Annual Meeting of the Society for American Archaeology, Albuquerque, NM.

Kenneth B. Vernon, Peter M. Yaworsky, Brian F. Codding

## 2018

### Prearchaic land use in Grass Valley, Nevada: This time with Bayes

Bi-annual Great Basin Anthropological Conference, Salt Lake City, UT.

Kenneth B. Vernon, Peter M. Yaworsky, Kate E. Magargal, D. Craig Young, David Zeanah, Brian F. Codding

**Prearchaic land use in Grass Valley, Nevada: A novel statistical implementation of optimal distribution modeling**

83th Annual Meeting of the Society for American Archaeology, Washington D.C..

Kenneth B. Vernon, Peter M. Yaworsky, Kate E. Magargal, D. Craig Young, David Zeanah, Brian F. Codding

**2015**

**Numic fires: Modeling the effects of anthropogenic fire on foraging decisions in the Great Basin**

80th Annual Meeting of the Society for American Archaeology, San Francisco, CA.

Kenneth B. Vernon, Kate E. Magargal, Ashley K. Parker, Will Rath, Brian F. Codding

## Teaching

**Teaching interests**. Behavioral Ecology, Urban Science, Environmental Data Science, Statistics, Geography, Conservation & Sustainability, Human Prehistory, North American Prehistory.

**University of Utah**

- ANTH 1010: Culture and the Human Experience
- ANTH 1030: Introduction to World Prehistory
- ANTH 5850: Quantitative Analysis of Archaeological Data

## Professional Affiliations

**Great Basin Anthropological Association** | 2016-present

**Society for American Archaeology** | 2014-present

**Coalition for Archaeological Synthesis** | 2022-present

**European Association of Archaeologists** | 2023-present

## Service, Outreach, and Other Activities

**Studio Lab** | University of Colorado, Boulder, 2022-2024.
**Mentor**. Supporting undergraduate research.

**Undergraduate Research Opportunities Program**. | University of Utah, 2018-2022.
**Mentor**. Supporting undergraduate research.

**Journal Club: Problems in Evolutionary Anthropology** | University of Utah, 2018-2022.
**Organizer**.

**Hopi Tribe v Donald J. Trump, Utah Diné Bikéyah v Donald J. Trump, Natural Resources Defense Council v Donald J. Trump** | 2018.
Brief Amici Curiae of Archaeological Organizations in Support of Plaintiffs.
**Contributor**.

## References

*Available upon request.*